CVL Assignment 04

Object Tracking: Template vs Optical Flow

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Class: CS

Dataset: MOT15 ADL-Rundle-6

1. Introduction

Object tracking is the task of estimating the position of a target across video frames after it has been localised in the first frame. This report evaluates two classical tracking approaches, Template-Based Tracking and Optical Flow-Based Tracking, applied to the ADL-Rundle-6 pedestrian sequence from the MOT15 benchmark. The sequence features a moving camera, moderate crowd density, and partial occlusions, making it a representative real-world test.

2. Dataset

ADL-Rundle-6 is part of the MOT15 benchmark. It is recorded from a moving camera in an outdoor pedestrian area. Ground truth bounding boxes and unique object IDs are provided per frame in the standard MOTChallenge format. We track Object ID 1 over 150 frames for evaluation.

Sequence: ADL-Rundle-6 Benchmark: MOT15 Camera: Moving Scene: Outdoor pedestrian area Frames evaluated: 150 Target: Pedestrian ID 1

3. Methods

3.1 Template-Based Tracking

A reference patch is cropped from the target bounding box in the first frame. In each subsequent frame, this patch is compared against a local search window around the previous position using Normalised Cross-Correlation (cv2.TM_CCOEFF_NORMED). The location with the highest similarity score becomes the new bounding box centre. The template is not updated, keeping the method simple but making it sensitive to appearance changes over time.

3.2 Optical Flow-Based Tracking (Lucas-Kanade)

Shi-Tomasi corner features are detected within the initial bounding box and tracked frame-to-frame using sparse Lucas-Kanade optical flow (cv2.calcOpticalFlowPyrLK). The bounding box displacement each frame is computed as the median of all successfully tracked feature displacements, making the method robust to outlier features and partial occlusion. When too few features remain, re-detection is triggered inside the current estimated box.

4. Evaluation Metrics

Both trackers are evaluated using IoU (Intersection over Union) against the ground truth bounding box per frame. Three summary statistics are reported.

Mean IoU is the average IoU over all frames. Higher values indicate better overall localisation.

Success Rate is the percentage of frames where IoU is at or above 0.5, representing the percentage of frames considered successfully tracked.

Average Speed is the wall-clock time in milliseconds per frame. Lower values indicate a faster tracker.

5. Results

Both trackers were applied to Object ID 1 over 150 frames of ADL-Rundle-6.

Template Matching: Mean IoU: 52.34% Success Rate: 50.00% Average Speed: 3.54 ms/frame Best frames: Early sequence Occlusion handling: Poor, drifts Appearance adaptation: None (fixed template)

Optical Flow (LK): Mean IoU: 39.75% Success Rate: 28.00% Average Speed: 4.67 ms/frame Best frames: Throughout sequence Occlusion handling: Fair, re-detects Appearance adaptation: Implicit (features update)

6. Analysis and Discussion

6.1 Template-Based Tracking

Template matching is straightforward to implement and performs adequately in the early frames of ADL-Rundle-6, where the pedestrian's appearance closely matches the initialisation patch. However, as the target moves through the crowd and undergoes partial occlusion, template drift becomes significant. Because the reference patch is never updated, small errors accumulate across frames. Additionally, scale and rotation changes are not handled, meaning the tracker degrades progressively in a moving-camera sequence.

6.2 Optical Flow-Based Tracking

Lucas-Kanade sparse optical flow operates on the actual pixel displacement between consecutive frames rather than a fixed reference, making it inherently more adaptive. The median displacement estimator filters outlier features caused by background motion or partial occlusion. Re-detection inside the estimated box partially addresses feature loss, though sudden large motion can still cause the tracker to drift. Per-frame computation is typically faster than template correlation over a large search window.

6.3 Limitations

Template matching has high sensitivity to appearance changes, drifts under occlusion, cannot handle scale change, and is limited by the search window under fast motion. It requires no training data.

Optical flow has low sensitivity to appearance changes, can re-detect features after occlusion, cannot handle scale change, and is prone to feature loss under fast motion. It also requires no training data.